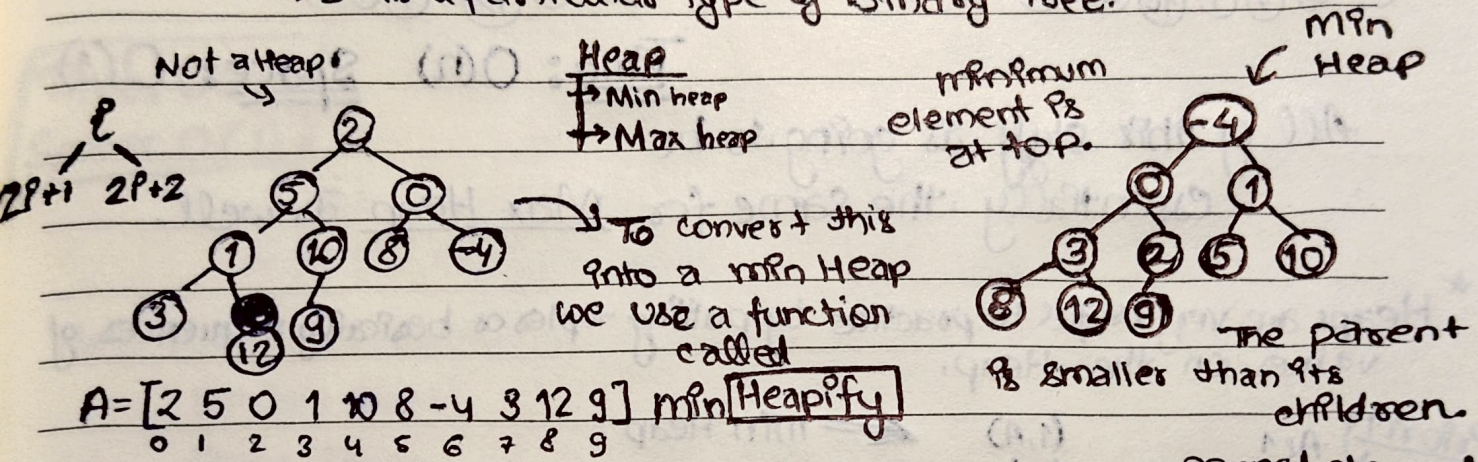
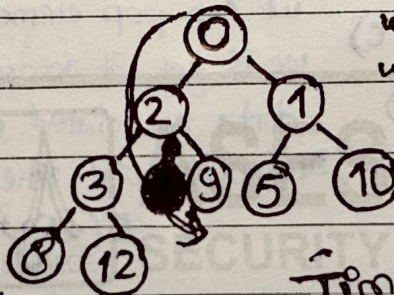


Heaps and Priority Queues

- ↳ these two things means exactly the same thing.
- ↳ It is a particular type of Binary Tree.



If we pop (4) from the min Heap then it'll have to fix itself so the min. element is at top.



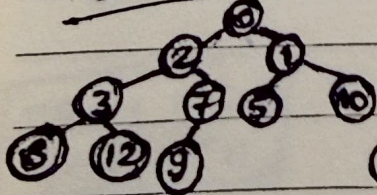
we swap the popped element with the smallest element [direct child] and keep swapping till the empty space falls up.

Time: $O(\log_2 n)$

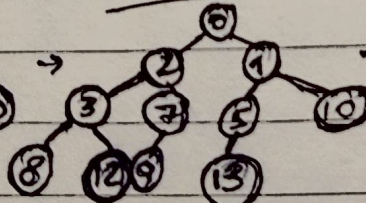
whenever we do stuff that's in relation to the height of the tree when the tree is roughly balanced height wise it takes $O(\log_2 n)$.

[[Heap pop (Extract min)]]
Time: $O(\log_2 n)$

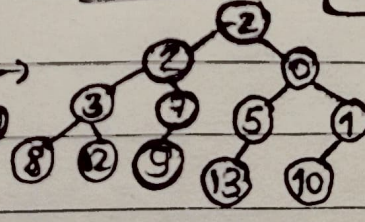
Insert 7



Insert 13



Insert -2



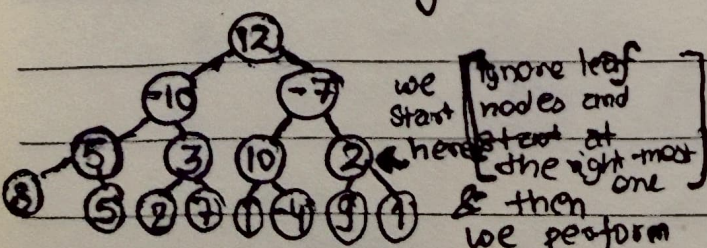
Heap peek: $O(1)$

Heap sort repeatedly pop off the minimum

Time: $O(n \log n)$
Space: $O(1)$

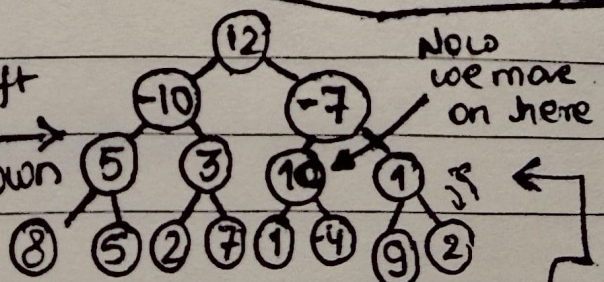
Heapify

Convert binary trees into Heaps



we start at the root at the right-most one & then we perform a function called **sift down**

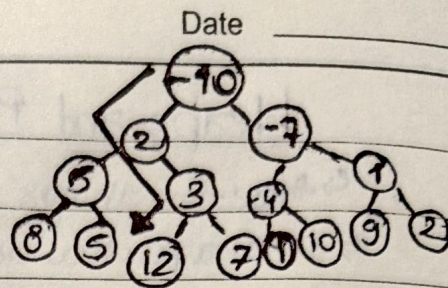
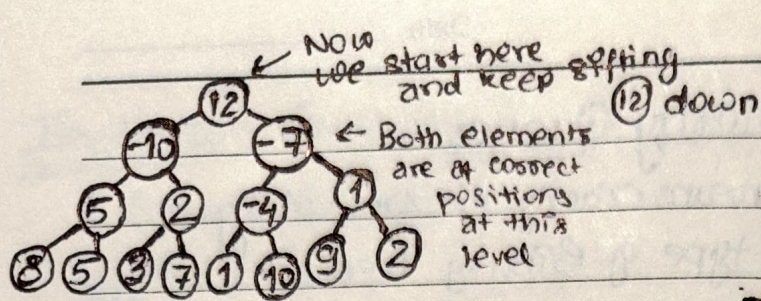
sift down



Now we move on here

keep sifting down at this level

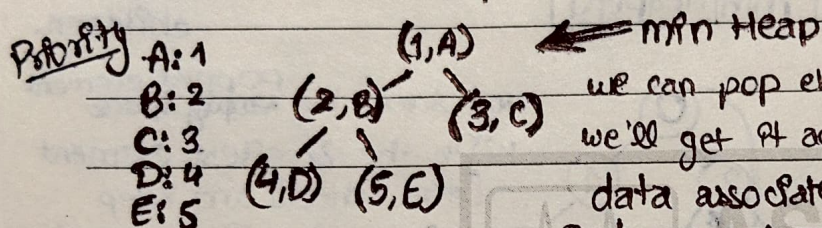
Putting (2) at its correct position



Time: $O(n)$ Space: $O(1)$

All of this stuff is going to be essentially the same for Max Heap as well.

* Heaps are very useful in practice by putting tuples or basically sequences of values on the Heap.



we can pop elements from this min Heap and we'll get it according to priority along with the data associated with it. and promote the rest upwards all in $O(\log n)$ time.